1 Architectures

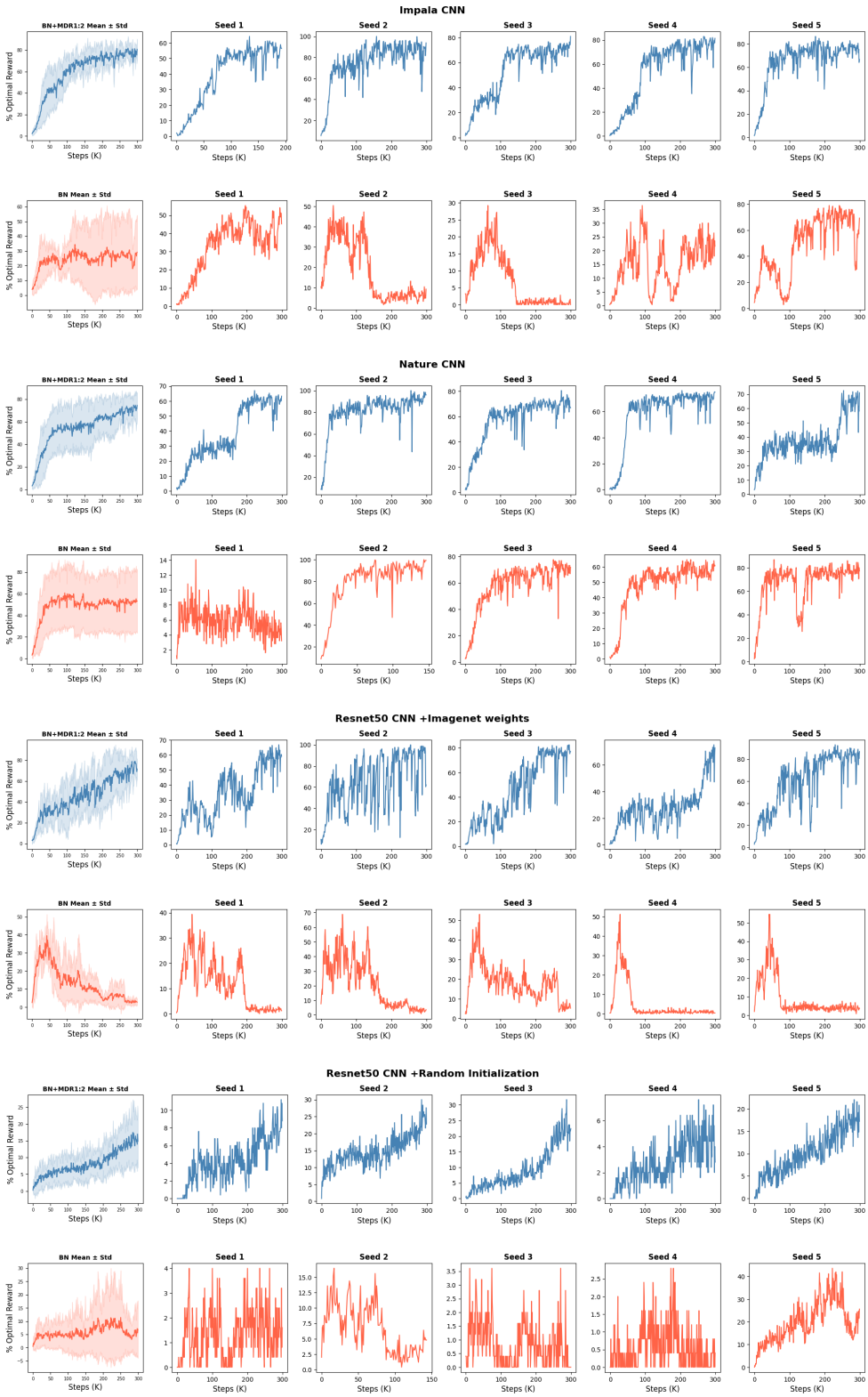


Figure 1: Reward curves across all tested architectures. Each block contains two rows: **BN** (Red) and **BN+MDR** (Blue). The first column shows the **mean** ± **std** across all seeds; subsequent columns show **individual seeds**. ResNet-50 with random initialisation exhibits early entropy collapse, preventing the policy from learning. Hence the LR ↓ and Entropy ↑ hyperparameters have been adjusted accordingly.

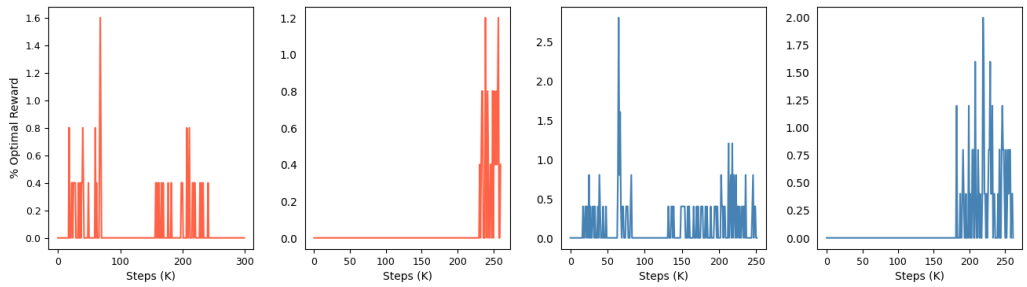


Figure 2: Resnet50 +Random initialization Trained with the same entropy and learning rate as the other architectures. Blue and Red curves follow the same structure as Figure 1

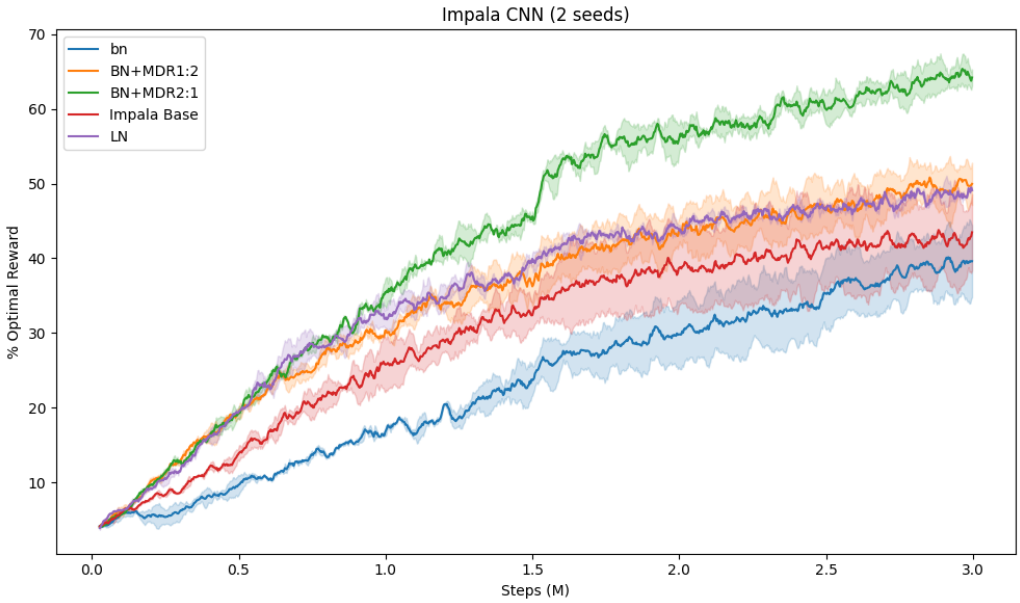
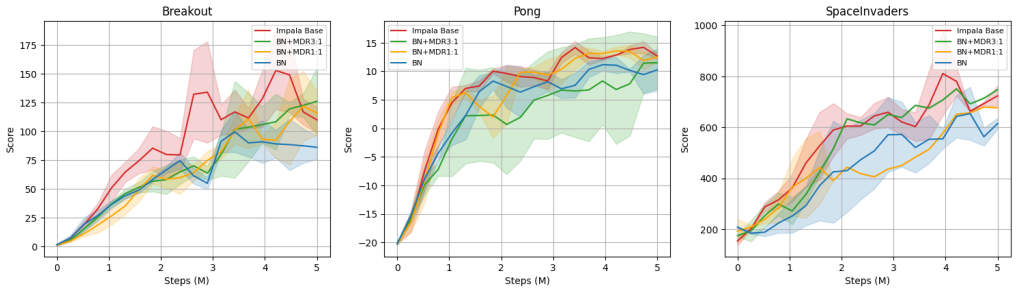


Figure 3: Impala CNN trained on the 256 Histopathology patch localization environments, Impala Base is the base Impala CNN architecture without normalisation, variants including BN and LN are presented

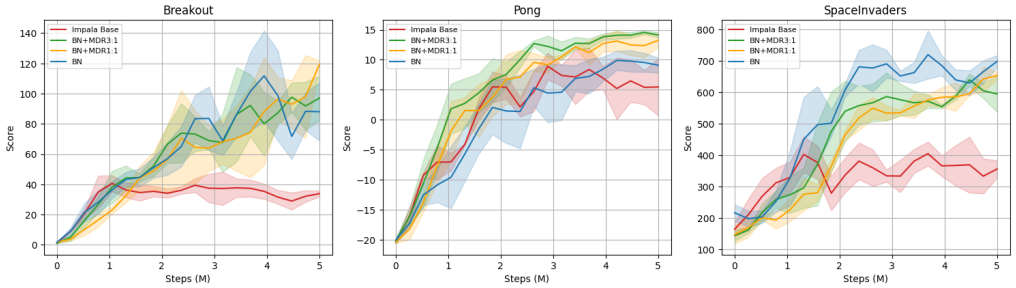
2 Atari

Method	Hyperparameters	Breakout	Pong	Space Invaders
Impala Base	Original	128.6 ± 51.5	11.2 ± 1.3	718.7 ± 48.2
	Aug. LR & Critic	33.7 ± 3.6	4.8 ± 4.0	410.8 ± 45.0
BN	Original	100.4 ± 27.9	10.7 ± 1.9	597.5 ± 74.0
	Aug. LR & Critic	94.5 ± 16.0	6.7 ± 4.5	666.4 ± 32.3
BN+MDR3:1	Original	131.2 ± 26.1	10.3 ± 6.0	698.4 ± 8.0
	Aug. LR & Critic	95.2 ± 26.4	13.2 ± 0.5	611.7 ± 21.2
BN+MDR1:1	Original	105.1 ± 19.4	11.6 ± 1.6	644.6 ± 41.2
	Aug. LR & Critic	98.7 ± 18.4	10.7 ± 2.1	626.3 ± 28.9

Table 1: Performance comparison across games (+ Sticky actions) and configurations after 5M steps. Each method is evaluated under the **original** hyperparameters and with **augmented learning rate and critic coefficient** (See Table 2 Red values).



(a) Original Hyperparameters



(b) Augmented learning rate and critic coefficient

Figure 4: Learning curves on Atari games. Solid lines show mean episodic return across 3 seeds. Shaded regions show standard deviation.

Table 2: PPO Hyperparameters for Atari Experiments

Hyperparameter	**Atari**
Environment	
Number of environments	8
Steps per environment	128
Frames per batch	1,024
Total frames	5M
Sticky actions	Yes
Optimization	
Optimizer	Adam
Learning rate	2.5×10^{-4} (0.001)
Adam ϵ	1×10^{-5}
Gradient clipping	0.5
Update epochs	4
Minibatch size	256
PPO	
Clip parameter ϵ	0.1
Entropy coefficient c_2	0.01
Value function coefficient c_1	0.5 (1.0)
Advantage Estimation	
Discount factor γ	0.99
GAE parameter λ	0.95
reward clipping	Yes
Architecture	
Encoder	Impala
Actor/Critic heads	Linear